

Understanding Uncertainty in Data-Driven Research

How Uncertainty Shapes Machine Learning



Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Aleatoric and Epistemic Uncertainty

Unit 3



Types of uncertainty

Aleatoric Uncertainty

Latin *alea*, game of dice, gambling

Inherent randomness in outcomes

Irreducible in principle



Epistemic Uncertainty

Greek *episteme*, knowledge

Lack of knowledge about models or processes

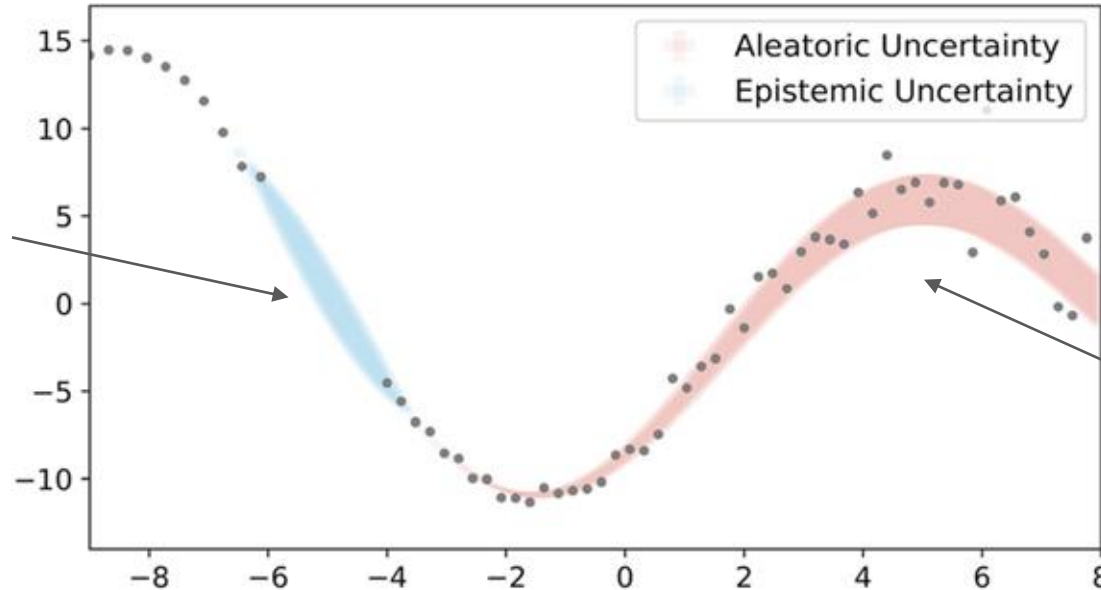
Reducible in principle



Types of uncertainty

Epistemic

No knowledge
about shape
of curve



Aleatoric
Remaining
variance

Common Assumptions

- Aleatoric = data uncertainty = irreducible
- Epistemic = model uncertainty = reducible
- Increase training data = reduce epistemic uncertainty^{1,2}

⇒ Definitions of types not consistently defined

⇒ Causes of uncertainty not related to types

¹ Eyke Hüllermeier and Willem Waegeman. 2021. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning*, 110(3):457–506.

² Alex Kendall and Yarin Gal. 2017. What uncertainties do we need in bayesian deep learning for computer vision? *Advances in neural information processing systems*, 30.

There is a close connection between **data** and **model**.

Sources of Uncertainty

Omitted Variables



missing features

Errors in X



imperfect observations

Errors in Y



incorrect labels

Deployment



changing processes

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Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Omitted Variables, Errors in X and Y, Deployment

Unit 3



Types of uncertainty

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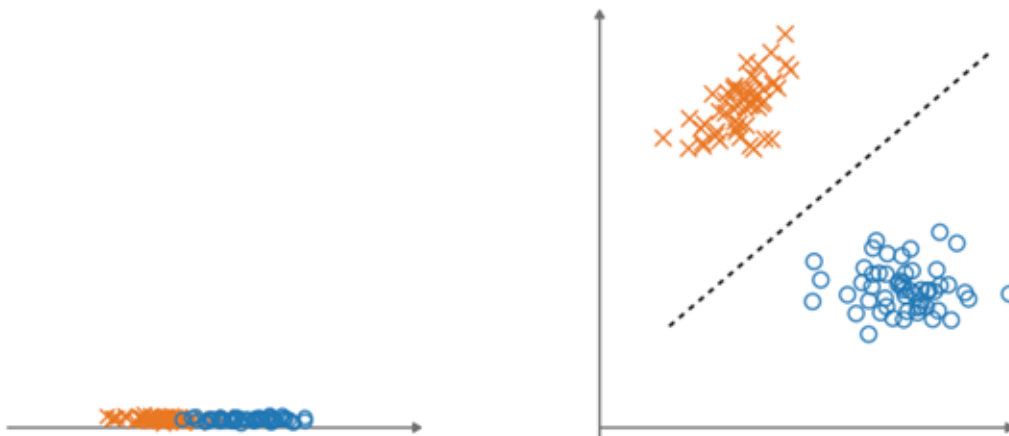
Deployment



changing processes

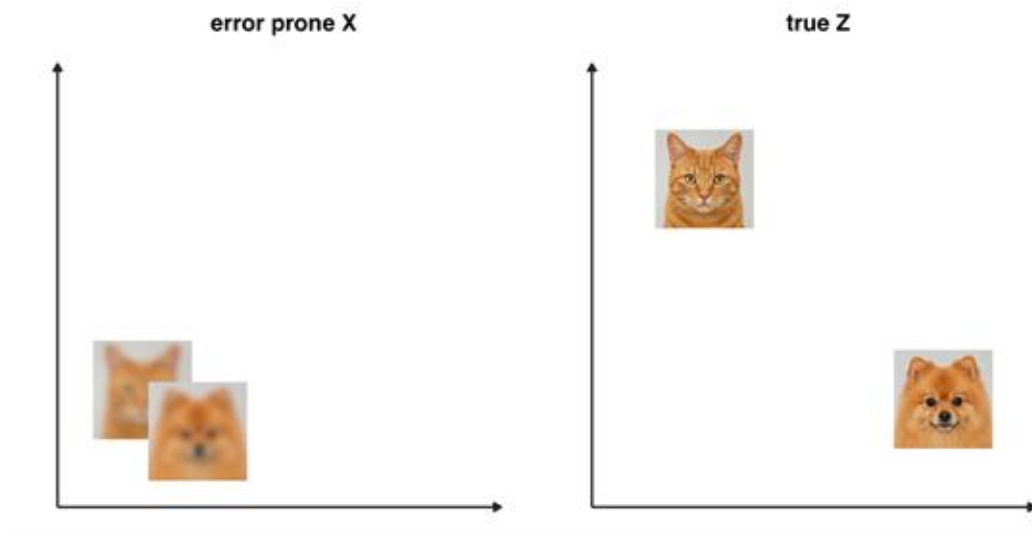
Omitted Variables

- How much information does the omitted variable bear about Y ?
- The larger the bias, the larger the uncertainty
- With additional features we can predict Y with less uncertainty



Errors in X

- Imperfect measurement (e.g., low-resolution image)
- Usage of proxy variables
- Subjectivity in data collection



Errors in Y

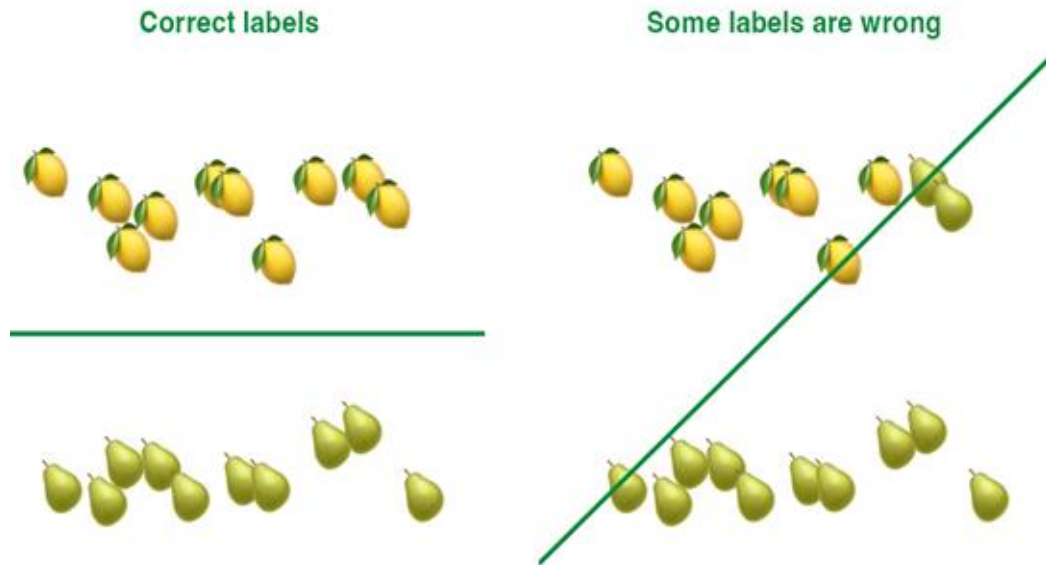
Noisy labels through

- annotation
- measurement
- processing

Even unsystematic errors lead to bias

Bias = model uncertainty = epistemic uncertainty

BUT cannot be reduced with increased sample size



What is reducible?

The definition of reducible uncertainty depends on which **actions** are available:

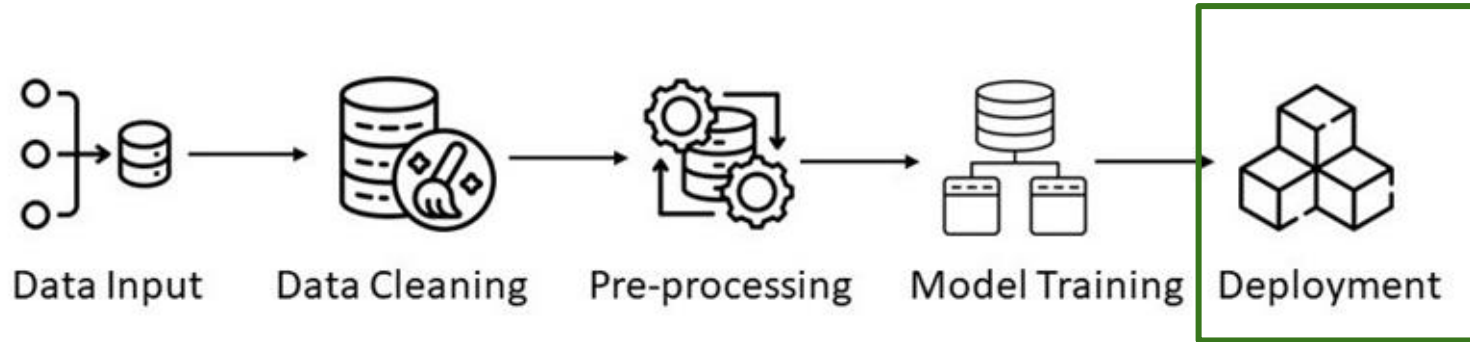
- Can I add features?
- Reduce feature noise?
- Reduce label noise?

Reducing these sources of uncertainty results in **reduced aleatoric uncertainty**



Data are always designed, deliberately or unintentionally, although the eventual data analyst might be uninvolved or even unaware of decisions going into the data-generating process.

Deployment



Are relations learned by the model still valid?

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Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Bayesian Neural Networks

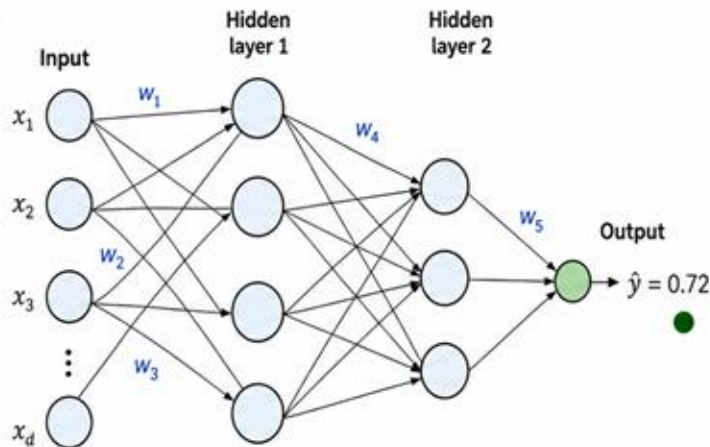
Unit 3



Bayesian Neural Networks

Standard neural network

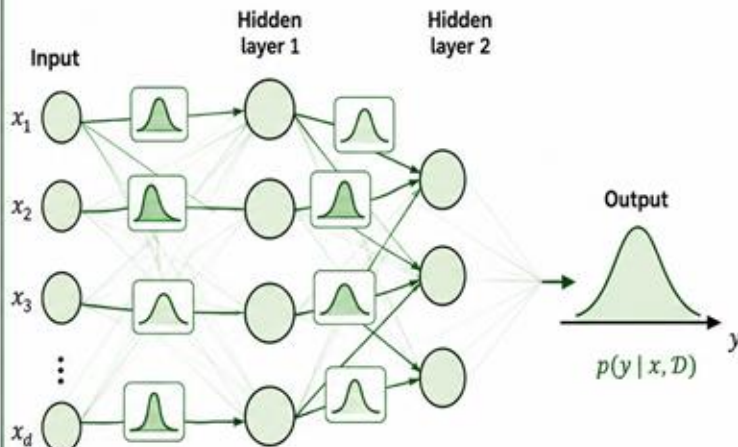
fixed weights



Learns **one** set of weights
Produces **one** point prediction

Bayesian neural network

distributions over parameters



Learns **distributions** over weights
Produces a **predictive distribution**

Bayesian Neural Networks



Bayes' rule gives the posterior distribution

$$p(w \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid w)p(w)}{p(\mathcal{D})}$$

$$\textit{posterior} = \frac{\textit{likelihood} \cdot \textit{prior}}{\textit{evidence}}$$

w weights of neural network

$\mathcal{D}$ training data

$p(w)$ prior over the weights

We effectively learn **a distribution over plausible neural networks.**

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Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Inference in Bayesian Neural Networks

Unit 3



Inference in BNNs

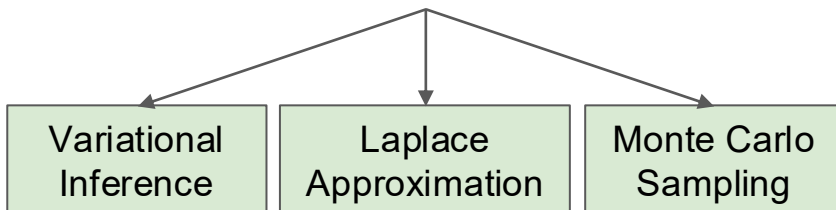
Highly non-linear functions



Exact inference intractable



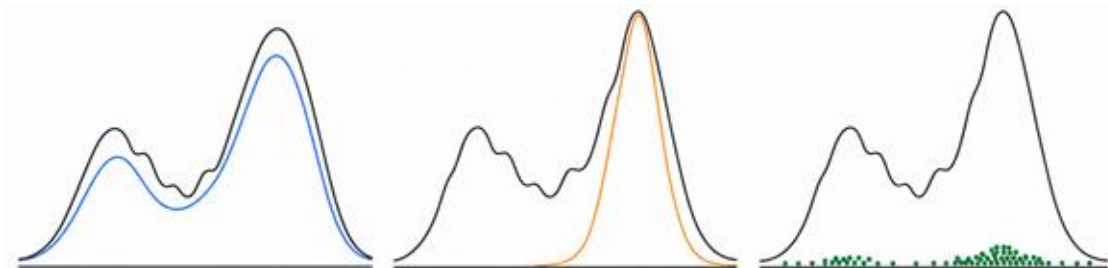
Approximate inference



$$p(w \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid w)p(w)}{p(\mathcal{D})}$$

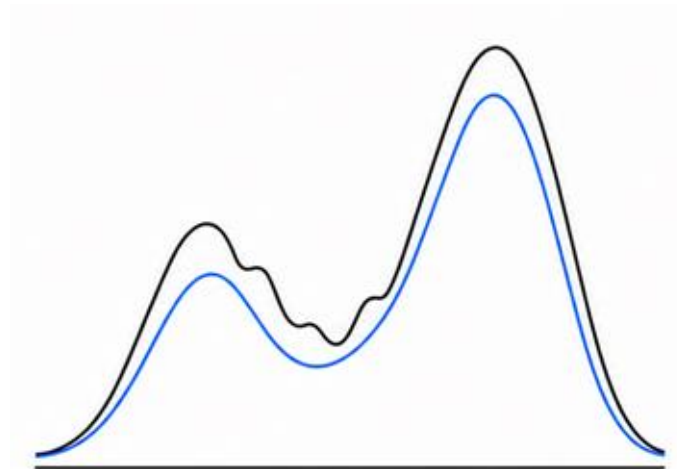


difficult



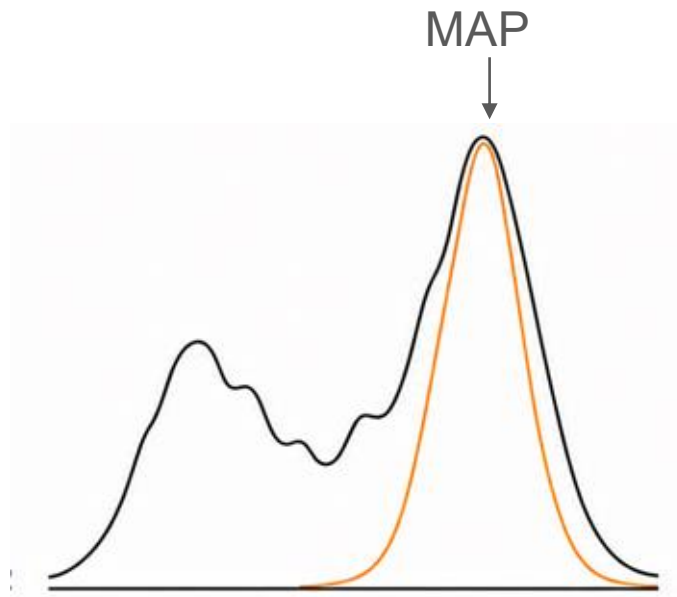
Variational Inference

- approximate true posterior with a family of simpler tractable distributions
- use e.g., a set of independent Gaussians
- find member that is closest to true posterior



Laplace Approximation

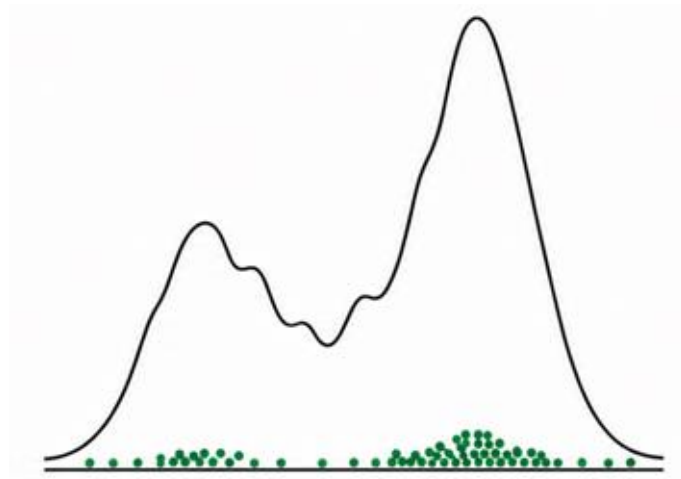
- approximate true posterior with a single Gaussian distribution
- centered at the distribution's mode (the Maximum A Posteriori, or MAP, estimate)



Monte Carlo Sampling

- approximate posterior by drawing random samples
- samples converge to desired distribution

Not to be confused with **Deep Ensembles!**



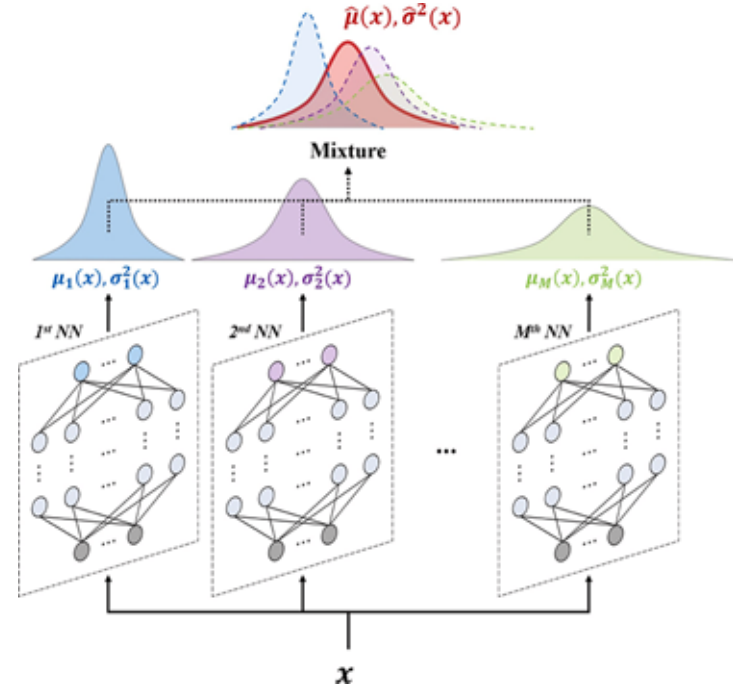
Deep Ensembles

Monte Carlo

sample from a single approximated or true parameter posterior distribution

Deep Ensembles

train multiple independent deterministic models from different random initializations to capture distinct modes in function space



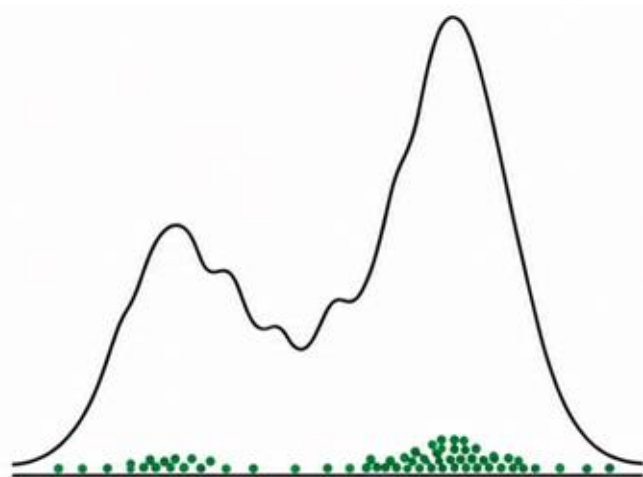
What does prediction look like?

For a **new observation** x^* , the Bayesian predictive distribution is

$$p(y^* \mid x^*, \mathcal{D}) = \int p(y^* \mid x^*, w) p(w \mid \mathcal{D}) dw$$

"Consider all plausible neural networks, weight their predictions by how plausible they are given the data, and average over them."

Monte Carlo Sampling



In practice: **approximate** the integral by drawing several weight configurations

$$w^{(1)}, \dots, w^{(T)} \sim p(w \mid \mathcal{D})$$

and making one prediction with each

$$p(y^* \mid x^*, w^{(1)}), \dots, p(y^* \mid x^*, w^{(T)})$$

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Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Quantifying Aleatoric and Epistemic Uncertainty

Unit 3



Quantifying Aleatoric and Epistemic Uncertainty



$$\underbrace{H[\mathbb{E}_w p(y \mid x, w)]}_{\text{total}} = \underbrace{\mathbb{E}_w [H(p(y \mid x, w))]}_{\text{aleatoric}} + \underbrace{I(y, w \mid x, \mathcal{D})}_{\text{epistemic}}$$

total: predictive entropy

aleatoric: expected entropy of the individual models

epistemic: mutual information between the prediction and the model parameters,
calculated as *epistemic* = *total* – *aleatoric*

Example 1: Strong disagreement

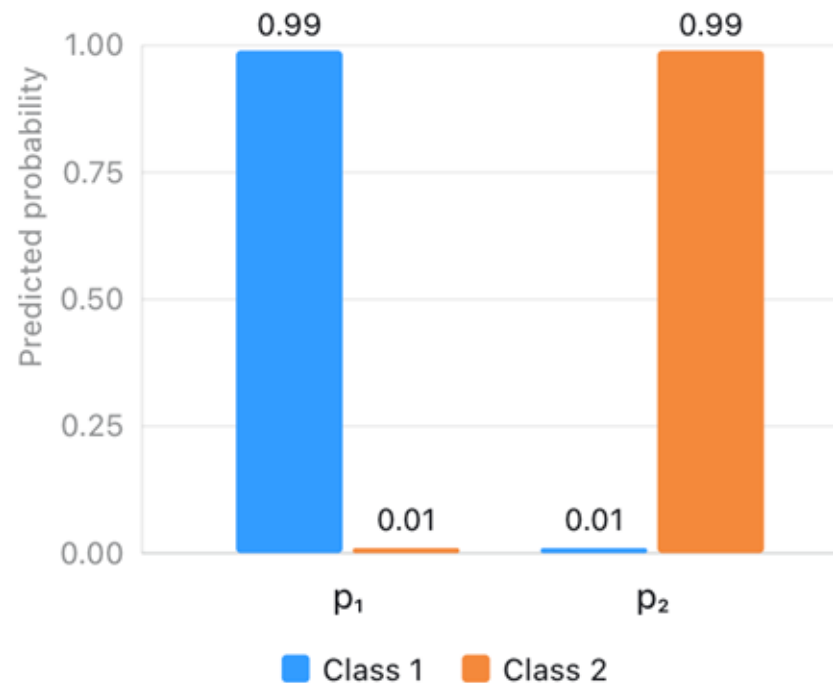
$$p_1 = (0.99, 0.01), p_2 = (0.01, 0.99)$$

The mean predictive distribution is

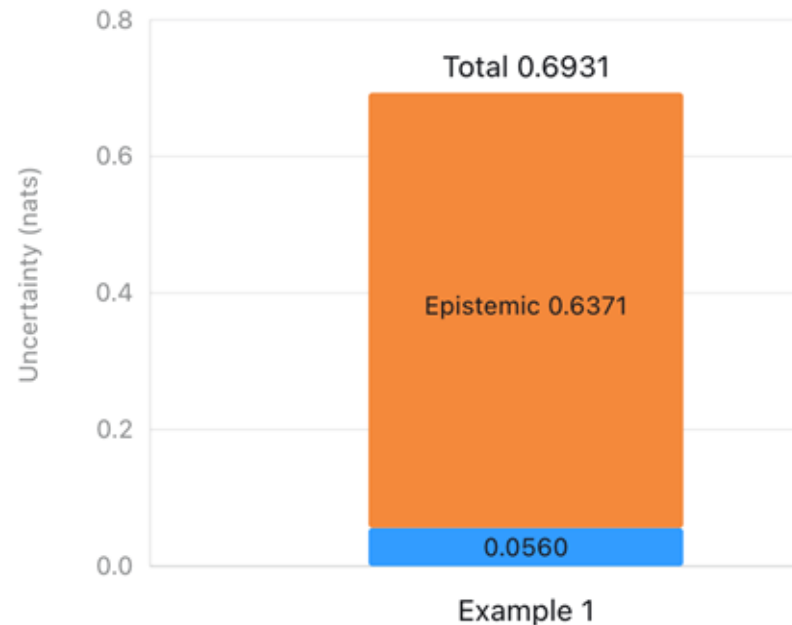
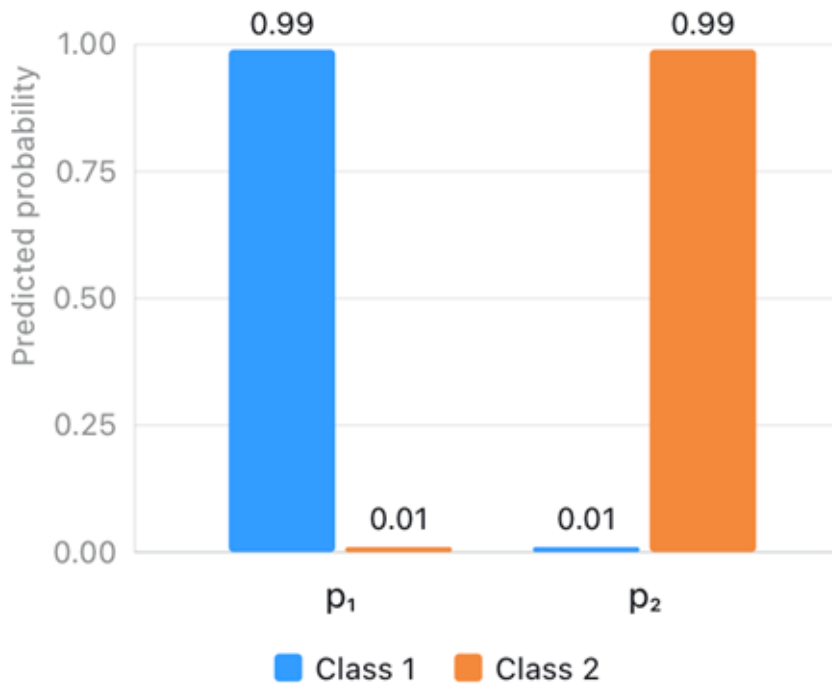
$$\bar{p} = \frac{1}{2}(p_1 + p_2) = (0.5, 0.5)$$

Total predictive uncertainty

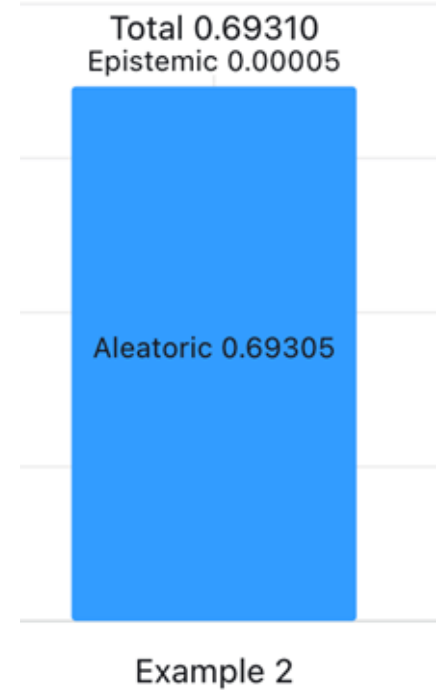
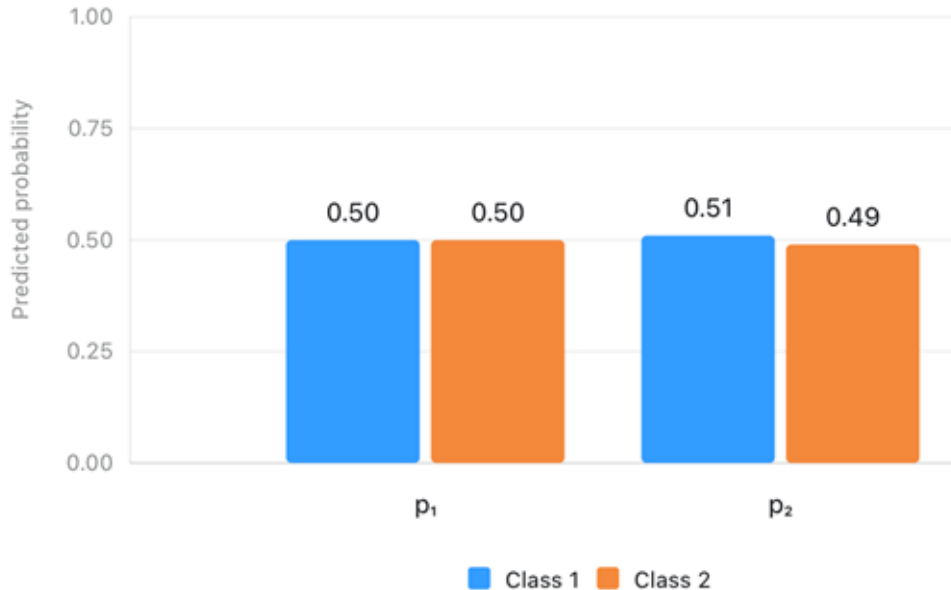
$$\begin{aligned} U_{\text{total}} &= H(\bar{p}) \\ &= -[0.5 \log(0.5) + 0.5 \log(0.5)] \\ &= 0.6931 \end{aligned}$$



Example 1: Strong disagreement



Example 2: Both uncertain



Comparison

Example 1

$$p_1 = (0.99, 0.01)$$

$$p_2 = (0.01, 0.99)$$

Example 2

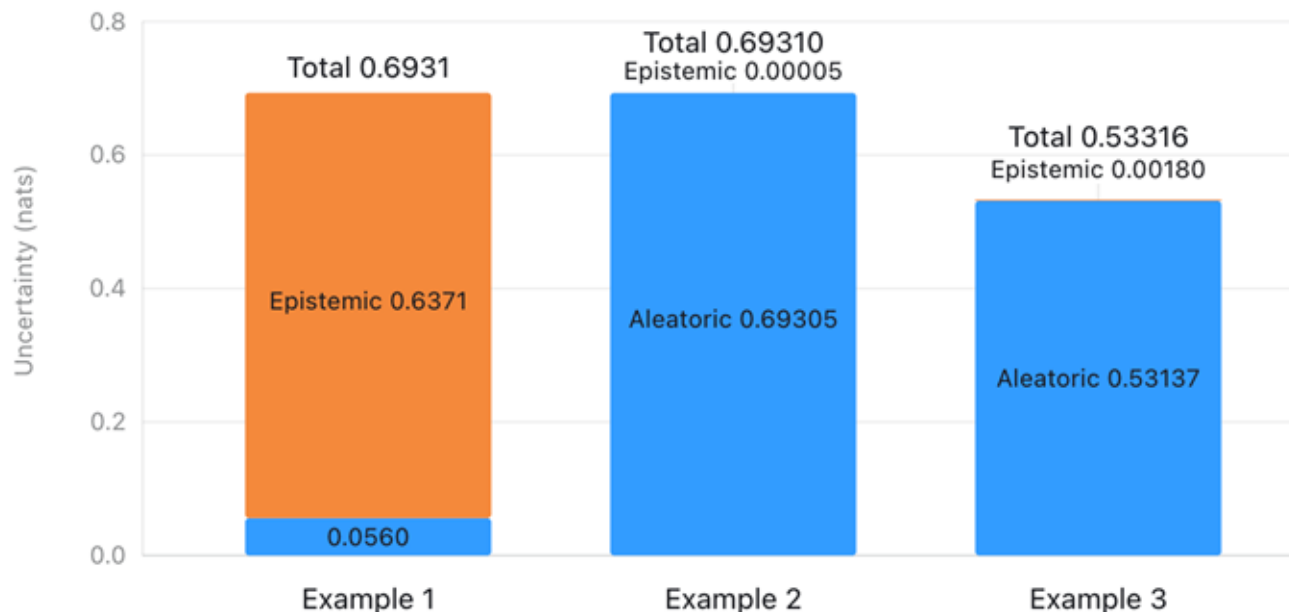
$$p_1 = (0.50, 0.50)$$

$$p_2 = (0.51, 0.49)$$

Example 3

$$p_1 = (0.75, 0.25)$$

$$p_2 = (0.80, 0.20)$$



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Understanding Uncertainty in Data-Driven Research

Sources of Uncertainty in Machine Learning Applications & Caveats of Uncertainty Decomposition

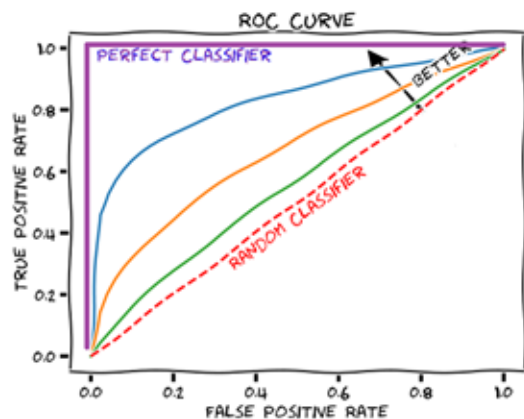
Unit 3



Applications predictive uncertainty

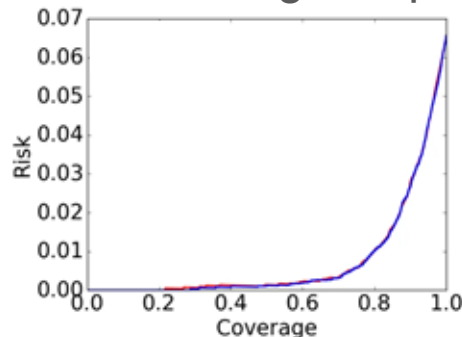
Correctness prediction:

Wrong predictions shall have higher uncertainties than correct predictions.



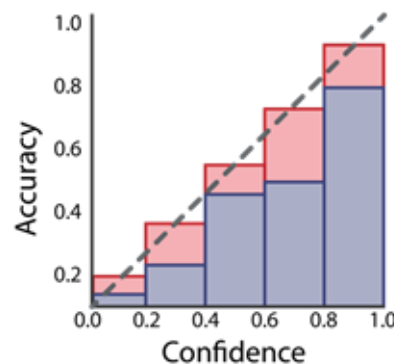
Abstained prediction:

Refuse to predict on the $x\%$ most uncertain examples. Then calculate the model's accuracy on the remaining samples.



Calibration:

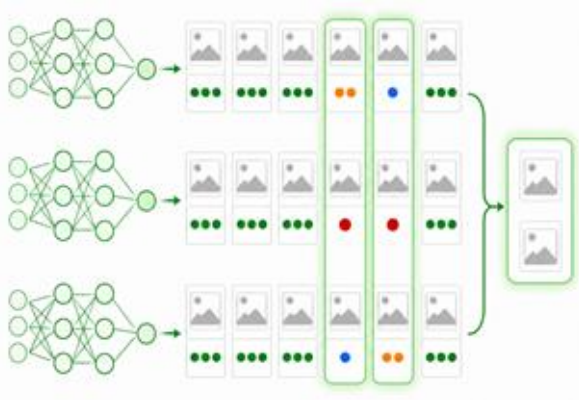
predicted probabilities shall match frequentist probabilities



Applications AU vs. EU

Active Learning:

BALD score maximizes EU



Out of Distribution:

OOD samples

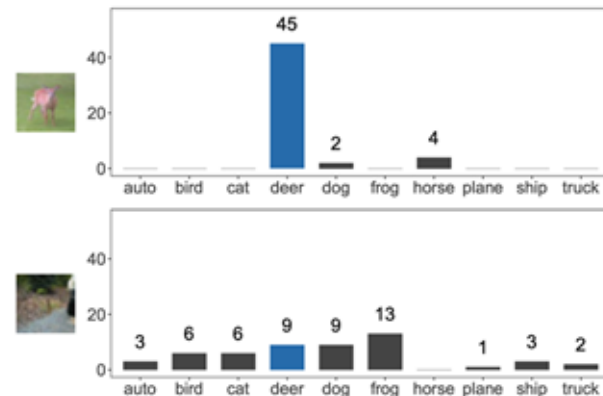
↔ high EU



Human Disagreement:

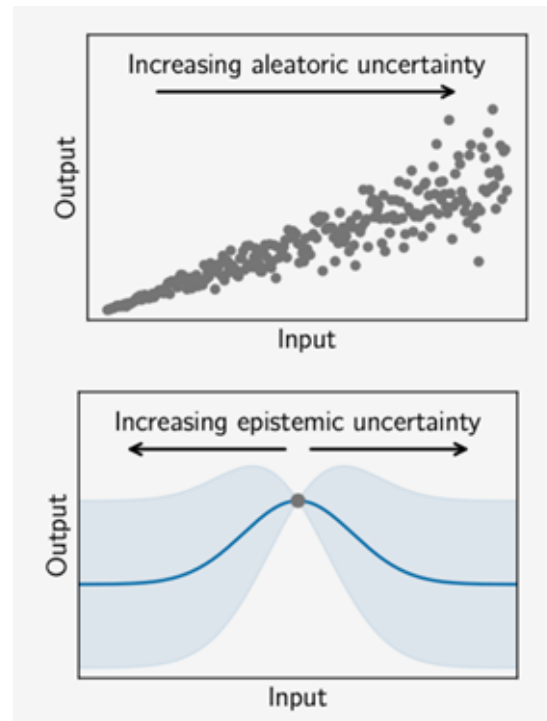
human disagreement

↔ AU



Caveats of AU vs. EU

- Quantifying AU and EU only makes sense if really needed for downstream task
- Correct AU estimation is only possible if EU is sufficiently low
- True quantities of interest almost always have to be approximated in the real world. Ignoring approximation errors can lead to dangerous assumptions and ineffective methods.



Caveats of AU vs. EU



Predictions
are conflated
with data-
generating
processes

Spurious
claims are
made about
decomposition
on unseen
data

Uncertainty
and accuracy
are
misleadingly
connected

Concepts
receive
multiple
distinct
mathematical
definitions

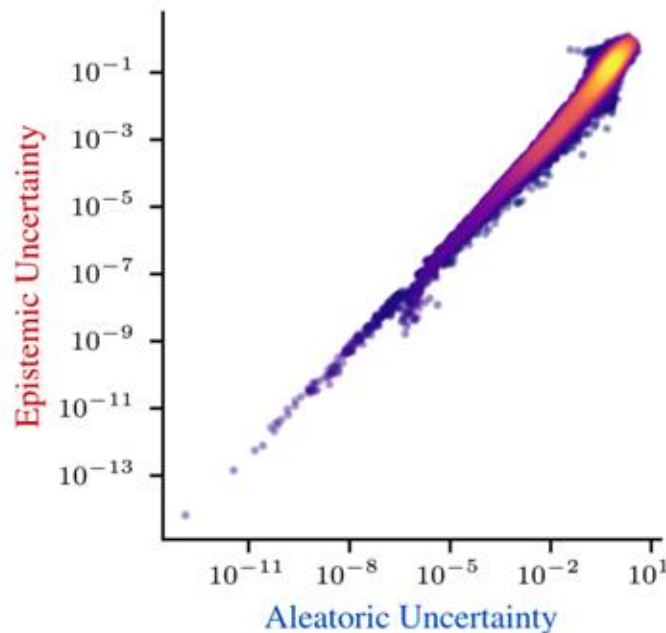
Estimates of AU and EU are correlated

Various uncertainty decomposition methods provide **correlated estimates for AU and EU**.

A "one-size-fits-all" uncertainty estimator does not exist.



Different tasks need different methods.



Goals vs. overloaded definitions

Measure a model's predictive uncertainty

Reason about uncertainty's reducibility

Assess a model's predictive performance

Describe statistical dispersion in data

Aleatoric uncertainty

"captures noise inherent in the data"

"cannot be reduced with more data"

Epistemic uncertainty

"uncertainty in the model parameters"

"can be explained away given enough data"

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